Danny Hu

danh295.github.io/Portfolio | linkedin.com/in/danny-hu | github.com/Danh295

Education

University of Waterloo & Wilfrid Laurier University — Waterloo, ON
Honours Bachelor of Computer Science & Bachelor of Business Administration Double Degree — *Sep 2023 – Present*

Technical Skills

Languages: JavaScript/TypeScript, Python, SQL (Oracle), C/C++, HTML/CSS
Frameworks/Libraries: React, Next.js, FastAPI, Node.js, Tailwind CSS, OpenCV, PyTorch, Hugging Face
Tools: Git/GitHub, Linux (Bash, Make, GDB, Valgrind), MongoDB, Figma, py-spy, line_profiler, psutil

Experience

Software Solutions Developer Co-op — Sep 2026 – Present
Independent Electricity System Operator (IESO) — *Mississauga, ON*

- Developing a **Python** performance monitoring wrapper on **psutil** for deployment on a Windows QA server, capturing process-level CPU and memory metrics from scripts without modifying their source.
- Black-box and integration tested **psutil**, **py-spy**, **line_profiler**, and **tracemalloc** against unmodifiable target programs, validating Windows compatibility under enterprise-approved package constraints.
- Benchmarked profiling overhead across tools, ranging from 0.5% (psutil) to 31.8% (tracemalloc) on tested workflows, and presented the trade-offs to the Solutions Development team, scoping psutil for the initial production rollout.
- Code-reviewed **Python** data pipelines and **Oracle SQL** queries for correctness, security, and maintainability.

Software Developer Co-op — Jan 2026 – Apr 2026
City of Waterloo — *Waterloo, ON*

- Independently built a full-stack web app to digitize handwritten parking tickets, achieving 92.3% per-field accuracy across 15 extracted fields per page on a 39-page labeled benchmark dataset.
- Designed a 10-stage computer vision pipeline in **Python** using **OpenCV** and **FastAPI**, combining grid-snap zone detection, contour-based checkbox analysis, and a plate OCR fallback merging **PaddleOCR** with a fine-tuned **TrOCR** model via character-level SequenceMatcher alignment.
- Fine-tuned Microsoft's **TrOCR-base-handwritten** on real plate crops using **PEFT/LoRA**, merging the adapter into the base weights at startup so inference runs with no added latency.
- Shipped a **Next.js**, **React**, **TypeScript** frontend with in-browser camera capture and an editable multi-page review, running fully local inference with no cloud APIs to meet municipal data-privacy requirements.

Web Developer Associate — Jan 2025 – Apr 2025
Civiconnect — *Lincoln, ON*

- Developed a travel-agency web app for Lincoln, ON in **React**, **Next.js**, and **Strapi**, featuring a **Google Places API** map, filterable events, user authentication, reviews, and a calendar-based itinerary builder.
- Delivered a client site for the Welland Law Association in **React**, **Tailwind CSS**, and **Next.js**, backed by a custom **Strapi** schema with search and filtering across site content.

Projects

Opera AI *(Hack Canada 2x Winner) | FastAPI, Python, Next.js, Gemini API* — Mar 2026

- Won $2,000+ across two tracks at Hack Canada 2026 with a multimodal appliance repair assistant; co-built the backend and AI workflow.
- Implemented **FastAPI** microservices that pass user-uploaded media to **Gemini** multimodal models and return step-by-step repair instructions.
- Integrated Cloudinary for media handling and ElevenLabs for voice output into the **Next.js** frontend within the 48-hour build window.

PuppetMaster *(DDC x Ignition Hacks 2nd Place) | React, Next.js, MediaPipe, HTML5 Canvas* — Apr 2026

- Won 2nd place with a browser-based animation studio that animates 2D characters from live webcam motion.
- Mapped 33 **MediaPipe** pose landmarks to joint rotations on a 2D puppet drawn in **HTML5 Canvas** at ∼30 fps.
- Kept **React** re-renders to ∼4/sec by holding per-frame pose data in refs outside the render cycle; added gesture-controlled recording, a trimmable timeline, and HTML export.